

ADVANCED COMPUTER PROGRAMMING TICT 2134(P)

Pharmacy Management System

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Project Introduction

The system manages important pharmacy functions such as medicine management, supplier management, stock monitoring, sales processing, user authentication, and sales reporting. It helps users track medicine details including price, quantity, and expiry dates, ensuring proper stock control and reducing losses due to expired medicines.

Role-based login functionality is implemented to provide secure access, where Admin users can manage medicines and view reports, while Staff users handle sales and supplier-related tasks. The sales report module calculates total income, tax, and final amounts, providing clear financial insights for decision-making.

Overall, this project provides a user-friendly, efficient, and reliable solution for pharmacy operations, making it suitable for small to medium-scale pharmacies.

Problem Statement

Manual pharmacy management systems are inefficient, error-prone, and time-consuming, making it difficult to track medicine stock, sales, expiry dates, and income accurately. An automated system is required to improve efficiency, accuracy, security, and reporting in pharmacy operations.

Team Members and Their

Registration Number	Name	Role
2022/ICTS/150	GNANESWARAN G.	Report writing, documentation
2022/ICTS/17	LAKSHITHA M.G.A.	Testing, debugging
2022/ICTS/73	RATHNAYAKA A.R.M.Y.N.	Event Handling & Validation
2022/ICTS/98	SAVISAHAN K.	System design, integration
2022/ICTS/123	HASEENA J.J.	Database, crud operations

Objectives of the System

- To automate pharmacy operations and reduce manual work.
- To manage medicine, stock, sales, and supplier records efficiently.
- To generate accurate sales reports with tax calculations.
- To provide secure, role-based access for Admin and Staff users.

Technologies Used

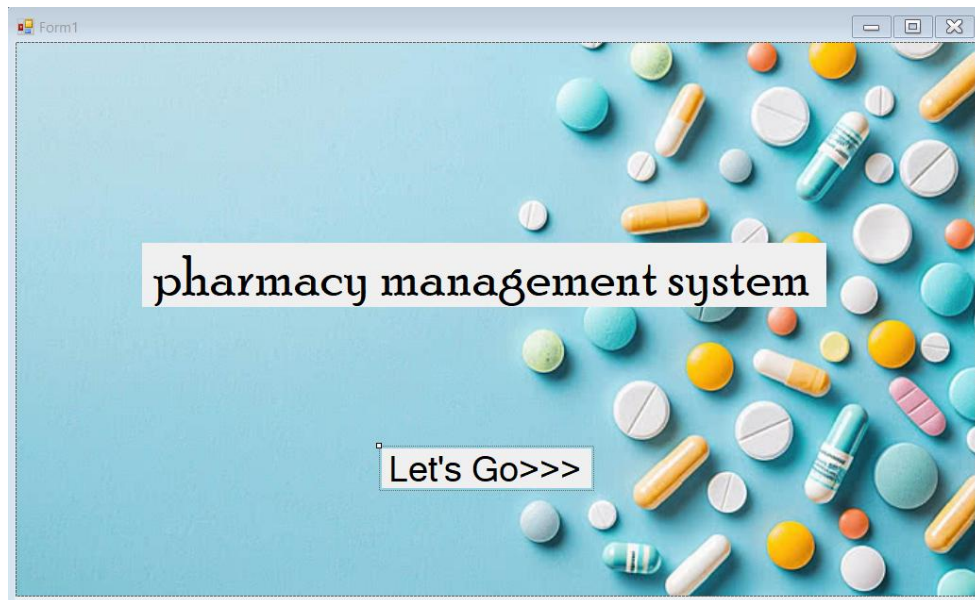
- Programming Language: C#
- Framework: .NET Framework
- IDE: Visual Studio
- Database: SQL Server

Form1

Purpose: To welcome users and start the system.

Function: Displays system title and navigates to the main interface.

Features: Labels, Button, Background Image, Event Handling, Windows Forms UI.



Form2 (Login Form)

Purpose

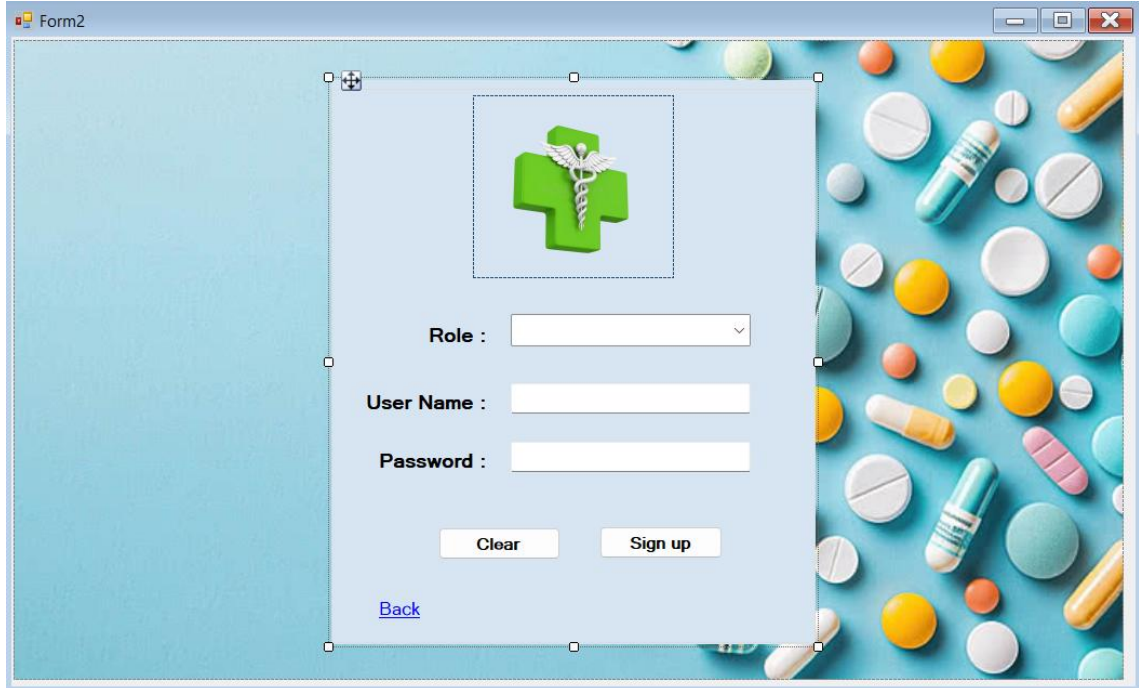
To facilitate User Registration and access control for a medical or pharmacy management system.

Function

- Account Creation: Registers new users with a unique username and password.
- Access Control: Assigns specific permissions based on the selected "Role."
- Input Management: Allows users to submit data or reset the form.

Features Used

- Dropdown Menu (Combo Box): For role selection.
- Input Fields: Text boxes for credentials.
- Action Buttons: "Sign up" to submit and "Clear" to reset.
- Navigation Link: A "Back" option to return to the previous screen.
- Themed UI: Medical iconography and a pharmaceutical background for context.



The screenshot shows a Windows-style window titled "Form2" with a light blue background. The window contains a registration form with a medical theme. At the top center of the form is a green 3D cross icon with a white caduceus. Below the icon are three input fields: "Role" (a dropdown menu), "User Name" (a text box), and "Password" (a text box). At the bottom of the form are two buttons: "Clear" and "Sign up". A blue underlined link labeled "Back" is located at the bottom left of the form. The background of the window is a collage of various colorful pills and capsules on a blue surface.

Form3 (Medicine Management Form)

Purpose

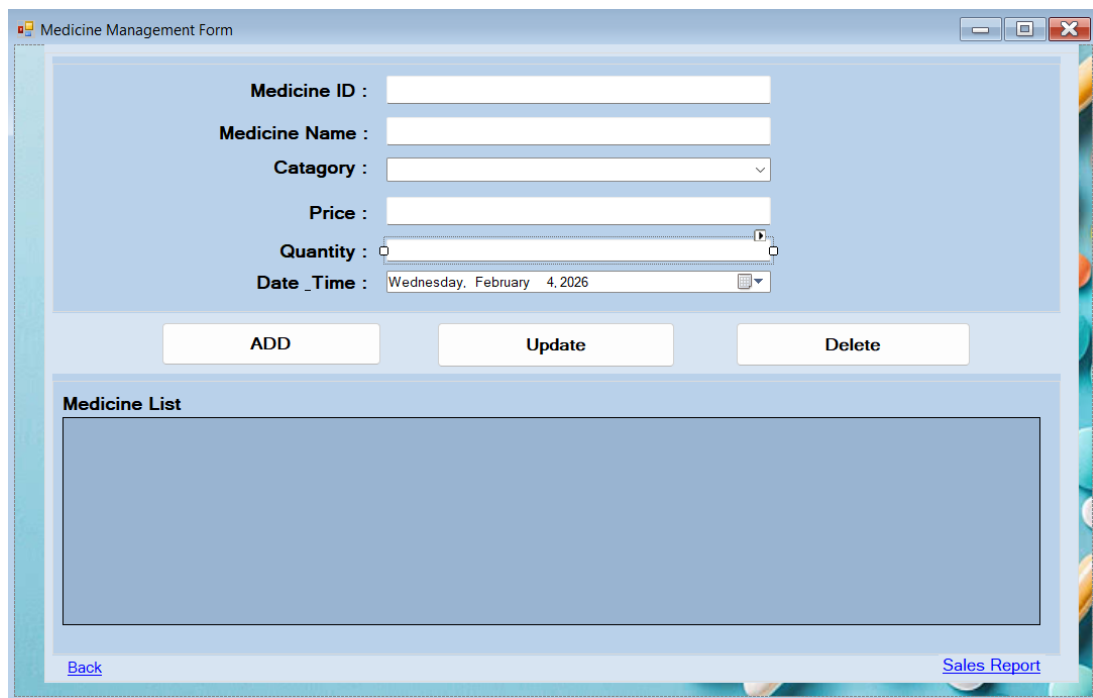
The Medicine Management Form is designed for Inventory Control, allowing staff to manage the digital database of pharmaceutical stock.

Function

- Inventory Maintenance: Supports CRUD operations (Add, Update, Delete) for medicine records.
- Stock Monitoring: Tracks specific quantities, pricing, and categories for each product.
- Navigation: Provides quick access to the "Sales Report" and the previous menu.

Features Used

- Form Controls: Text fields (ID, Name, Price) and a Dropdown for Categories.
- Specialized Inputs: A Numeric Spinner for Quantity and a Date Time Picker for scheduling.
- Data Visualization: A Data Grid View (Medicine List box) to display existing records.
- Action Buttons: Large "ADD," "Update," and "Delete" buttons for database management.



The screenshot displays a web application window titled "Medicine Management Form". The interface includes several input fields: "Medicine ID" (text), "Medicine Name" (text), "Category" (dropdown), "Price" (text), "Quantity" (numeric spinner), and "Date_Time" (datetime picker showing "Wednesday, February 4, 2026"). Below these fields are three buttons: "ADD", "Update", and "Delete". A section titled "Medicine List" contains a large, empty rectangular box, likely intended for a data grid. At the bottom left is a "Back" link, and at the bottom right is a "Sales Report" link.

Form4 (Supplier Management Form)

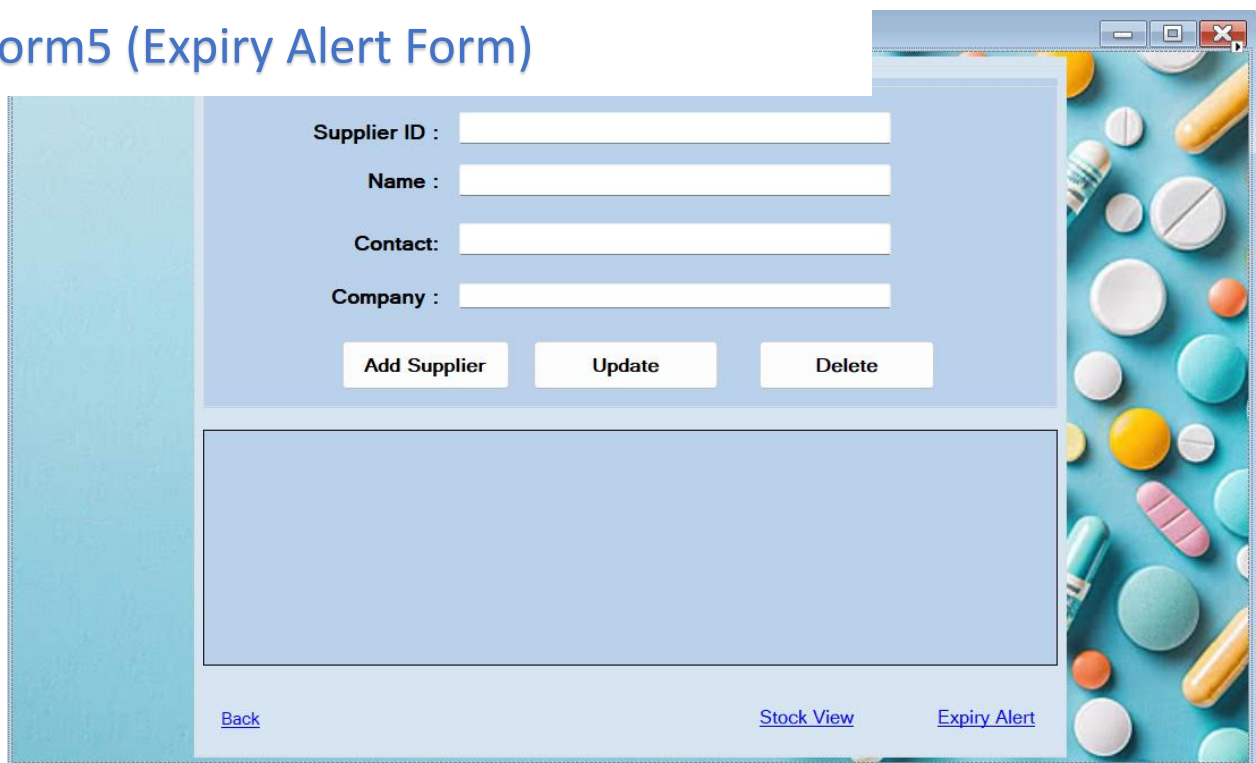
Purpose

To manage a database of vendor information, ensuring the pharmacy can easily contact and track companies that provide their medical stock.

Function

- Supplier Directory: Stores details like Supplier ID, Name, Contact info, and Company.

Form5 (Expiry Alert Form)



Supplier ID :

Name :

Contact:

Company :

[Back](#) [Stock View](#) [Expiry Alert](#)

Purpose

To monitor and manage expiring pharmaceutical stock. It serves as a safety and compliance tool to ensure expired medicines are removed from inventory promptly.

Function

- Expiry Tracking: Displays a list of medicines approaching or past their expiration dates.

Expiry Alert Form

Refresh :

Refresh

Back

Refresh

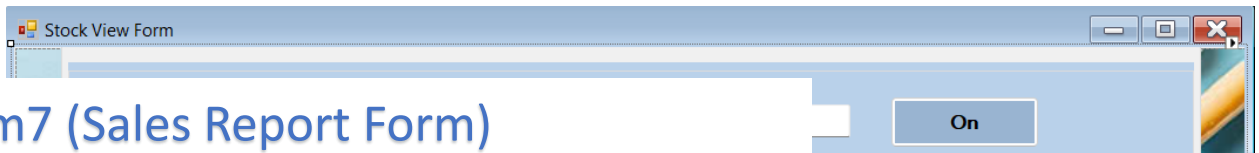
Form6 (Stock View Form)

Purpose

The main purpose is Inventory Inquiry, allowing users to quickly check the availability and total count of specific medicines within the system.

Function

- Stock Search: Enables users to look up inventory levels by "Medicine Name".



Form7 (Sales Report Form)

Purpose

To provide Financial Oversight and data-driven insights by summarizing sales performance and calculating tax obligations.

Function

- Report Generation: Allows users to filter and view specific types of sales data through a selection menu.
- Automated Calculation: Automatically computes Total Income, applies a 5% Tax rate, and determines the final Total Amount.
- Output Management: Enables users to produce physical or digital copies of financial data for record-keeping.

Features Used

- Dropdown Menu: "Select Report Type" for filtering data categories.
- Data Grid/Display Area: A central panel for viewing itemized sales records.
- Calculation Labels: Dedicated fields for displaying real-time financial totals (Income, Tax, Amount).
- Action Buttons: "Generate Report" to pull data and "Print Report" for final output.

Navigation: A "Back" button to return to the main dashboard or management forms.

Sales Report

Select Report Type :

Generate Report

Total Income : price

Tax(5%) : price

Total Amount : price

Print Report

Database Design

The Pharmacy Management System uses a relational database designed in Microsoft SQL Server. The database structure is normalized to reduce data redundancy and ensure data integrity

Navigation Flow



Conclusion

The Pharmacy Management System successfully provides an efficient and reliable solution for managing pharmacy operations. By using C# .NET and SQL Server, the system automates key processes such as medicine management, stock control, sales handling, and report generation. It reduces manual effort, minimizes errors, and improves data accuracy.

The role-based access system ensures secure usage for Admin and Staff users, while the sales reporting module helps in monitoring income and tax calculations. Overall, this project demonstrates how a well-designed software application can improve productivity and decision-making in a real-world pharmacy environment.